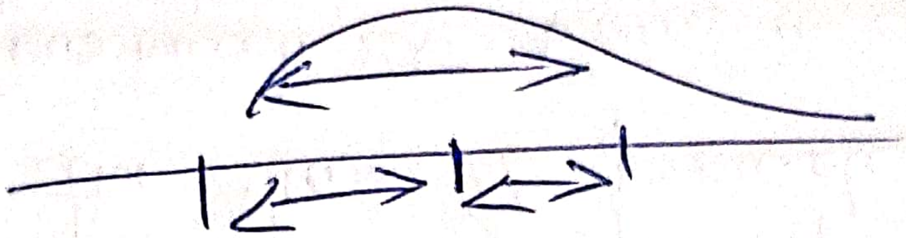
Measure of Dispersion 2nd vedio

Measure of dispersion describe the Spread or Variability of a dataset. They Indicate how much the Values in a dataset from the Central tendency.

Common Measures

1. Range
2. Variance
3. Standard deviation
4. Interquartile Range (IQR)

$x = \{1, 2, 3, 4, 5, 6, 7, 8)$



① Range

Def :- Range is the difference between the maximum value and minimum value in a datasets.

Range = Maximum Value − Minimum Value

Eg:- Ages {14, 13, 10, 20, 25, 75, 15}

Range = 75 − 10 = 65

Chars:

① Simple to Calculate

② Sensitive to outliers

③ Rough Measure of dispersion

$x = \{35, 40, 45, 34, 30, 70\}$

Rang = 40-30 = 10

Ran = 70-30
= 40

② Variance.

Def:- Variance Measures the average Squared deviation of Each Value from the Mean. It provides a Sense of how much the Values in a dataset Vary

Population Variance

$$\sigma^2 = \sum_{i=1}^{N} \frac{(x_i - \mu)^2}{N}$$

$x_i \rightarrow$ Data point

$\mu \rightarrow$ Population mean

$N \rightarrow$ Population Size

Sample Variance

$$s^2 = \sum_{i=1}^{n} \frac{(x_i - \bar{x})^2}{n-1}$$

$x_i \rightarrow$ Data points

$\bar{x} \rightarrow$ Sample mean

$n \rightarrow$ Sample Size

Example :- Size of flower Petals

$\{5, 8, 12, 15, 20\} \Rightarrow$ Variance of the distribution

$$\sigma^2 = \sum_{i=1}^{N} \frac{(x_i - \mu)^2}{N}$$

$$\mu = \frac{5+8+12+15+20}{5}$$

$$= \frac{60}{5} = 12$$

$$\text{Variance} = \frac{(5-12)^2+(8-12)^2+(12-12)^2+(15-12)^2+(20-12)^2}{5}$$

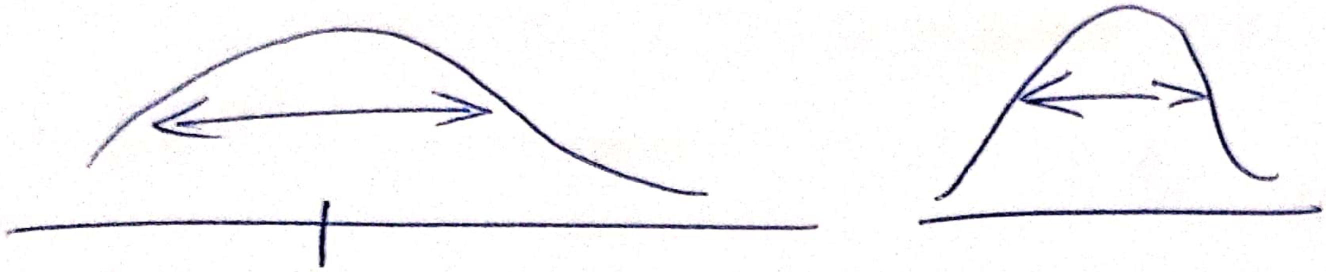
$$\text{Variance} = 27.6$$

Charteristics

① It provides a presis Measure of variability

② units are Squared of original

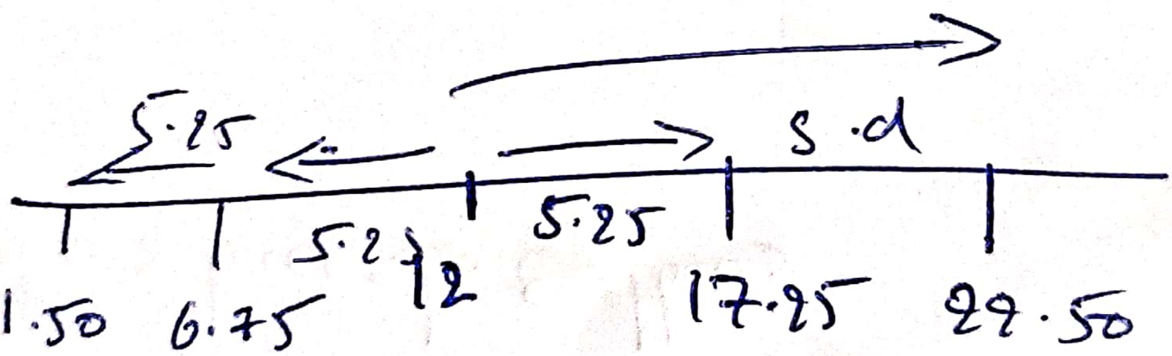
③ Sensitive to outliers of Dataset.



③ Standard deviation

Def :- The standard deviation is the square root of the variance

$$\sigma = \sqrt{27.6} \approx 5.25$$



$\{5, 8, 12, 15, 20, 24\}$

?

Z ~~test~~ score
(upcoming video)

↳ Standard normal distribution.

Characteristics

- It provides a clear Measure of Spread in the Same unit as the data.
- Sensitive to outliers